the day for them.

9 **(a)** packet of energy of electromagnetic radiation B1

energy = hf B1

Comments: A lot of candidates have missing keywords such as "electromagnetic radiation" and "Energy = hf". Another common mistake is that a number of candidates wrote "electromagnetic waves" which is not suitable in this context where photon is a particle and not a wave.

(b) **(i)** arrow below axis and pointing to right B1

(ii) 1 change in energy of photon = $\frac{hc}{\lambda_f} - \frac{hc}{\lambda_i}$ M1

$$= (6.63 \times 10^{-34} \times 3.0 \times 10^{8})\left(\frac{1}{6.84 \times 10^{-12}} - \frac{1}{6.50 \times 10^{-12}}\right)$$

$= -1.52 \times 10^{-15}$ J (accept +ve value) A1

(ii) 2 gain in kinetic energy of electron = loss in energy of photon

$\frac{1}{2} m v^2 = 1.52 \times 10^{-15}$

$\frac{1}{2} (9.11 \times 10^{-31}) v^2 = 1.52 \times 10^{-15}$ C1

$v = 5.78 \times 10^{7}$ m s^{-1} A1

(iii) conservation of momentum in either x or y direction C1

$$\left(\frac{6.63\times10^{-34}}{6.50\times10^{-12}}\right)=\left(\frac{6.63\times10^{-34}}{6.84\times10^{-12}}\right)\cos\theta+(9.11\times10^{-31})(5.78\times10^{7})\cos70°$$

OR

$$\left(\frac{6.63\times10^{-34}}{6.84\times10^{-12}}\right)\sin\theta=(9.11\times10^{-31})(5.78\times10^{7})\sin70°$$ C1

Solving: θ = 30.7° A1

Comments: Conservation of momentum is needed to be awarded marks

(c) (i) hf = Φ + KE_{max}

$$\frac{6.63\times10^{-34}\left(3\times10^{8}\right)}{6.84\times10^{-12}}=(180\times10^{3}\times1.6\times10^{-19})+KE_{max}$$ M1

KE_{max} = 2.79 x 10^{-16} J A1

Comments: This part is well done

(ii) $KE_{max} = eV_S$

2.79 x 10^{-16} = (1.6 x 10^{-19}) V_S M1

V_S = 1.74 kV A1

Comments: This part is well done

(iii) These electrons are emitted from the surface where least energy is required for emission (hence the remaining kinetic energy is maximum)

A1

Comments: This part is marked leniently. Candidates should improve on their answers even though they are awarded credit.

(d) (i) Energy needed to transit to level 2 = −0.46 x 10^{-15} – (–1.84 x 10^{-15}) C1

= 1.38 x 10^{-15} C1

Remaining energy = 1.52×10^{-15} – 1.38 x 10^{-15} = 0.14 x 10^{-15} J

= 1.4 x 10^{-16} J A1

(ii) The atom will not transit to higher energy level A1

because the energy of the photon is not equal to the difference between any two energy levels M1

Comments: This part is well done